

Event Density Theory

Primitives	P01	Existence layer (the foundational relational existence; below it, no further substrate structure)	Primitive
Primitives	P02	Participation as the primitive relation (chains participate — derived composite structures)	Primitive
Primitives	P03	Channel + locus indexing; spatial homogeneity (bundles three commitments, splittable)	Primitive
Primitives	P04	Bandwidth — a non-negative additive scalar	Primitive
Primitives	P05	Polarity-transport between adjacent loci	Primitive
Primitives	P06	Spatial dimension $D = 3+1$	Primitive
Primitives	P07	Channel structure as an ontological primitive	Primitive
Primitives	P08	Substrate scale I_{ED} (= I_P empirically)	Primitive
Primitives	P09	$U(1)$ -valued polarity	Primitive
Primitives	P10	Rule-type primitive (licenses V5 as a P10 kernel rule-type)	Primitive
Primitives	P11	Commitment — irreversible, with environmental phase-randomization (THE arrow primitive)	Primitive
Primitives	P12	Stability landscape $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$	Primitive
Primitives	P13	Time homogeneity; primitive event-discreteness (NOT 'the tick')	Primitive
Constants-inherited	Paper_012 (P-RB-1)	c — the rate of becoming (tick-to-hop conversion)	Constant
Constants-inherited	Paper_027	$G = c^3 \cdot I_P^2 / \hbar$ — Newton's constant	Constant*
Constants-inherited	Paper_029	$a_0 = cH_0/(2\pi)$ — the MOND transition scale. **2026-09-04: the $1/(2\pi)$ is NOT established as forced** — it cancels in Paper_028 §6.3 / Paper_029 §5.1's own displayed algebra and is reinstated by a normalization fixed nowhere. Re-tiered Postulated/disputed (gravity Staleness #10). UNAFFECTED: $a_0 \sim cH_0$, now reached FOUR independent ways, and $a_0(z) = cH(z)/(2\pi)$ exponent 1	Constant* — SCALE ok, COEFFICIENT DISPUTED 2026-09-04
Constants-inherited	Paper_038_5	Λ — the cosmological constant	Constant*
Constants-inherited	Paper_087	$\hbar$ — the action of one commitment (the grain of memory)	Constant
Constants-inherited	Paper_087 (P08) / 089	$I_P = I_{ED}$ — the grain (size of one hop)	Constant
Constants-inherited	cosmology	H_0 — the Hubble rate	Constant*
Constants-inherited	matter sector	Particle masses	Constant
Constants-inherited	matter sector	Gauge couplings g	Constant
Constants-inherited	matter sector	$\alpha \sim 1/137$ (fine-structure); mass ratios	Constant

Cosmology	SCBU	a0 & xi_canonical FORCED to share one substrate-cosmology boundary origin. **CHECKED 2026-09-04: survives — the shared object is R_H, not the coefficient; the 1/(2pi) dispute does not reach it, and a0 ~ cH0 gained a fourth route the same day. Tier unchanged. Residuals: xi_canonical is canon-INTERNAL and memory-sourced (Foundations staleness #4), and SCBU step 12 leaves xi_canonical (H0) OPEN — so 'FORCED' is conditional on P-Substrate-Cosmology-Unified. See Core Theory row 24.**	Postulated
Entanglement	063	E-1 factorizable $\Psi^{AB} = \Psi^A \times \Psi^B$ & the tensor-product structure	Postulated
Entanglement	064	V5-bilocal has Schmidt rank>1 generically	Postulated
Entanglement	065	Monogamy budget cap; measure saturates the budget	Postulated
Entanglement	069	ED sits at Tsirelson; PR-boxes excluded	Postulated
Gravity	035	Six acoustic-metric guardrails (C1-C6)	Postulated
Gravity	QuadraticStrain	P-Quadratic-Strain (Model C)	Postulated
QFT	019	P-YM-Action-Coarse-Graining	Postulated
QFT	020	P-OS-Reflection-Positivity (OS3)	Postulated
QFT	021	P-Profile-Rescaling (YM continuum survival)	Postulated
QM-kinematics	002	Tensor-product composition $\Psi^{AB} = \Psi^A \times \Psi^B$	Postulated
QM-kinematics	012	P-RB-1 local rate of becoming; substrate-c constancy	Postulated
Relativistic-QM	114	P-Gauge (paper-specific postulate)	Postulated
Substrate-evaluation	QuantumDarwinism	P-QD-LiveWeight (uncommitted branch-basis correlation consumes pairwise V5 weight)	Postulated
Substrate-evaluation	V5AttractiveSign/Relational Tick/MassWithoutMass	V5 existence as a cross-chain kernel	Postulated

Event Density Claims

Black-hole	039	Horizon = $b \rightarrow 0$ decoupling surface (Γ_{cross} collapse)	D-via-I / Form-forced
Black-hole	040	Trans-Planckian resolution (modes terminate at w_c)	D-via-I / Form-forced
Black-hole	041	Planck-mass remnant $M^* = c \cdot I_P$ (Scenario C)	D-via-I / Form-forced
Black-hole	042	No-singularity (C_{cum} bounded, $r \geq I_P$)	D-via-I / Form-forced
Black-hole	043	Area-law FORM $S \propto A$	D-via-I / Form-forced
Black-hole	044	BHPT scattering / Regge-Wheeler form	D-via-I / Form-forced
Black-hole	045	Helicity preserved (Schwarzschild) / amplified (Kerr)	D-via-I / Form-forced

Black-hole	046	Kerr twist $g_{t\phi}$	D-via-I / Form-forced
Black-hole	047	Hawking temperature $T_H = \kappa/2\pi$ via V5 KMS. [2026-09-04: this 2π is NOT the disputed one. It comes from near-horizon Euclidean smoothness (no conical defect) in Paper_BH_Thermal2Pi and does not cancel. The a_0 dispute (row 17) is about the azimuthal-Fourier factor in Paper_029. Whether they are the SAME 2π is exactly Route 2 of research target #18, and is open]	D-via-I / Form-forced
Black-hole	047	Non-thermal Hawking correction $[1 - c_1(w/w_c)^2]$	D-via-I / Form-forced
Black-hole	048	Scenario C forced (A, B excluded)	D-via-I / Form-forced
Black-hole	049	PBH relic abundance $\Omega_{\text{PBH}} \propto n_{\text{PBH}} \cdot M^*$	D-via-I / Form-forced
Black-hole	050	Page-curve rise + turnover + decline	D-via-I / Form-forced
Dark-sector	RelicLagrangian	Relic $\rho \propto a^{-3}$ from the P11 commitment number	Derived
Dark-sector	RelicLagrangian	Sector is TWO-COMPONENT, not one substance (L_{self} coarse-grains CANONICAL)	Derived
Dark-sector	RelicLagrangian	Pressureless-dust limit $w \rightarrow 0$ (fast-oscillating massive scalar)	Derived
Entanglement	064	Schmidt rank $> 1 \Rightarrow$ non-factorizable	Derived
Entanglement	065	Monogamy (CKW-form inequality)	Derived
Entanglement	066	No-signaling $p(a x,y)=p(a x)$, over-determined by 3 locks	Derived
Entanglement	068	von Neumann entropy form $S = -\text{Tr } \rho \log \rho$	Derived
Foundations	089	Theorem N1 — V1 finite-width (G bounded, decaying, scale l_{ED} ; delta-function and inf-width limits excluded)	Derived
Foundations	089/093	Theorem T18 — V1 strictly retarded support (the kernel-level arrow); advanced-V1 refuted by P11, symmetric-V1 non-constructible	Derived
Foundations	089/093	Wheeler-Feynman engagement: symmetric electrodynamics is mathematically consistent but substrate-operationally non-constructible in ED	Derived
Foundations	090/094	V5 inherits retardation from V1 (via T18); primitive-forward-causality - $>$ phenomenological arrows inherit under DCGT	Derived
Gravity	025	Holographic bound $N(R) \leq 4\pi R^2/l_{\text{ED}}^2$ (area law as edge-count)	Derived
Gravity	026	Newton $a_N = GM/R^2$ from the cumulative-strain reading	Derived
Gravity	032.5	Free chains follow acoustic-metric geodesics	Derived
Gravity	034/036	Deep-MOND $a = \sqrt{a_N a_0}$; flat-background AQUAL field eq	Derived
Gravity	037	a_0 continuum-limit invariant (does not RG-run)	Derived

Gravity	GR-I	Temporal lapse $N^2 \sim b \rightarrow$ weak-field Einstein metric (Schwarzschild branch, not Nordstrom)	Derived
Gravity	GR-I	Emergent $g \sim 1/b$ + factor-of-two light bending	Derived
Gravity	GR-II	Field-equation form $G + \Lambda \cdot g = \kappa \cdot T$	Derived
Gravity	GR-II	Matter conservation $\text{div } T = 0 \Leftrightarrow$ geodesic motion	Derived
Gravity	GR-II	ED gravity is KHRONOMETRIC (2 tensor + 1 khronon); gauge = foliation-preserving diffeos	Derived
Gravity	GR-II	$c_T = c$ structurally (GW170817 as identity); universal (not differential) Lorentz violation	Derived
Gravity	GR-III	$\alpha=1$ (P-Commitment-Linear) FORCED; Nordstrom excluded	Derived
Gravity	GR-III	Khronon scalar speed $c_s = c$	Derived
Gravity	GR-IV	$\alpha_2 = 0$ EXACTLY	Derived
Gravity	GR-IV	$\alpha_1 = -4 \cdot \lambda_{\text{local}}$; commitment FORCED sparse $\Rightarrow \lambda_{\text{local}} \ll 1$ (≥ 70 orders safe)	Derived
Gravity	MetricFromGraph	Reach law $p=1/(d-1) \Rightarrow g \sim 1/b$ in EXACTLY 3D	Derived
Q-compute	054	UR-1.1/1.2/1.3 three unresolvedness statements ($M_{\text{eff}} < M_{\text{cap}}$; $\Gamma_{\text{cross}} > \Gamma_{\text{decoupling}}$; $\Gamma_{\text{commit}} < \Gamma_{\text{indiv}}^{-1}$)	Derived
Q-compute	054/056	Existence of Γ_{commit} , $\Gamma_{\text{indiv}}^{-1} \sim c/l_{\text{ED}}$; $M_{\text{cap}} = (\Gamma_{\text{indiv}}/\Gamma_0)^{1/\gamma}$ as the UR-1.1 threshold	Derived
Q-compute	058b	$\alpha_{\text{topological}} = 0$ (per-locus): the 058 §3.3 saturation never triggers for topological codes	Derived
Q-compute	058b	Syndrome records are budget-free (P11 commitments)	Derived
QFT	016	Continuum gauge-covariant derivative $D_\mu = d_\mu + iA_\mu$	Derived
QFT	017	Lorentz covariance of the continuum free-scalar equation	Derived
QFT	019	Continuum YM equations $D_\mu F = J$ (Euler-Lagrange)	Derived
QFT	021	Spectral-gap FORM $\Delta \sim 1/l_*$ (Yang-Mills mass gap)	Derived
QFT	022	Substrate quotient = physical-state space; BRST = coarse-grained quotient	Derived
QFT	024	Lindblad/GKSL master-equation form	Derived
QFT	MS-II	Gauge group $U(N) = SU(N) \times U(1)$ from channel multiplicity	Derived
QM-kinematics	001	Complex polar carrier $P_K = \sqrt{b_K} \cdot e^{i\pi_K}$	Derived
QM-kinematics	009	Berry phase γ_C = holonomy of the coarse-grained P05 connection	Derived

QM-kinematics	010	Aharonov-Bohm phase from T17 bundle circulation	Derived
QM-kinematics	011	Bloch form $\psi_k = e^{i k x} u_k(x)$ from discrete translation symmetry	Derived
QM-kinematics	012.7	Galilean adjacency/propagation bandwidth-asymmetry	Derived
Relativistic-QM	103	Cl(3,1) frame uniqueness up to similarity	Derived
Relativistic-QM	104	Anyon prohibition in 3+1D	Derived
Substrate-evaluation	A1 (CommonCauseNotChannel)	Controlled channel capacity between distinct regions is EXACTLY zero; boundary = common-cause not channel	Derived
Substrate-evaluation	B4 (ChargeAsTopology)	Winding w in Z conserved + irreversibility-protected; orientation-blindness realizes the skeleton, withholds the field	Derived
Substrate-evaluation	CleanSubstrateVector	The parity-clean substrate is VECTOR for every channel-count ($\det H(k)$ even $\Rightarrow$ winding=0 forall N); parity violation NECESSARILY spontaneous	Derived
Substrate-evaluation	QuantumDarwinism	Committed records cost no V5 budget (accounting theorem) $\Rightarrow$ classical objectivity unbounded	Derived
Constants-inherited	Constants-paper §1	Constants ledger: $c/\hbar/l_P$ present at first events (dimensionful, ~units); $a_0/\Lambda/H_0$ /masses inherited late (dimensionless, the personality)	Grounded
Constants-inherited	Constants-paper §3	$G = c^3 \cdot l_P^2 / \hbar$; G absent from the SM; $[G_d] = m^d$ (units know the dimension count)	Grounded
Constants-inherited	Constants-paper §4	The dimension-signature in $G (l_P^{d-1})$ IS the holographic area-law / reach-law road — NOT a new independent road to 3D	Grounded
Cosmology	Cos-01	Inflation = post-SCBU substrate SATURATION regime; $a(t) = a_0 e^{\{Ht\}}$	Grounded
Cosmology	Cos-05	Dark energy = late-universe SATURATION regime; $w = -1$ (leading order)	Grounded
Cosmology	Cos-05	Λ -smallness (120-OOM) reduces to Route A + Friedmann; naive $\rho_{vac} \sim M_P^4$ STRUCTURALLY ABSENT substrate-side (no infinite mode-tower)	Grounded
Cosmology	DF2/DF4	Low-accel galaxies CONSISTENT WITH (not counterexamples to) derived a_0 , via commitment-density saturation suppression	Grounded
Cosmology	SCBU	a_0 & $\chi_{canonical}$ = two projection MECHANISMS of one boundary $R_H = c/H_0$	Grounded
Dark-sector	DarkSector	Committed NEUTRAL RELIC candidate for clusters/CMB	Grounded
Dark-sector	RelicLagrangian	Field content $\Phi = \sqrt{\rho/m_R} e^{i \theta}$ = coarse-grained neutral-channel participation measure	Grounded
Dark-sector	RelicLagrangian	Khronometric kinetic form L_{kin} (time-along- u + spatial c_{perp})	Grounded

Dynamics	Dyn-01	Saddle-Hessian-signature dynamics = the foundational generator; 3 regime classes (S1 collapse / S3 inflation-DE / S2 horizon-radiation)	Grounded
Dynamics	Dyn-02	Horizon-motion law: RDE($w=1/3$), MDE($w=0$), LDE($w=-1$) via substrate Noether stress-energy + DCGT + Friedmann	Grounded
Dynamics	Dyn-03	Radiation law: EM Larmor + GW quadrupole via Noether flux + DCGT	Grounded
Dynamics	Dyn-04	Gravitational collapse = monotonic Hessian-signature evolution toward all-compression; horizon = closed signature-class boundary	Grounded
Dynamics	Dyn-05	Compact-binary inspiral = two-saddle coupled evolution + quadrupole radiation reaction; Peters timescale	Grounded
Dynamics	GW-00	GW = propagating saddle-Hessian-signature reconfiguration (V1 retarded); energy = local ED-gradient load (no pseudotensor)	Grounded
Dynamics	GW-00	Quadrupole source-amplitude FORM $P_{GW} \sim \langle Q\text{-triple-dot}^2 \rangle$	Grounded
Dynamics	GW-01	BH ringdown = decay of a near-horizon mixed-signature saddle; mode quantization from Hessian eigenstructure	Grounded
Dynamics	GW-02	Stochastic background $\Omega_{GW}(f)$ = cosmologically-integrated NoetherFlux over saddle transitions	Grounded
Entanglement	063	E-2 bilocal cross-chain term Δ_{KL} exists	Grounded
Gravity	032	Six ED-10 weak-field prerequisites fulfilled at form level	Grounded
Gravity	033	Scalar-tensor acoustic-metric field eq (covariantization)	Grounded
Gravity	038.5	$\Lambda = V1$ cosmological-scale integral over R_H ; Λ CONSTANT, $w=-1$ forced, = frozen saturation-floor	Grounded
Gravity	KM-I	The khronon's deep field IS the MOND sector (unifies the two gravity programs)	Grounded
Gravity	KM-I	Lensing tracks the MOND potential, no added vector; no new ghost; PPN-safe	Grounded
Gravity	KM-II	Orthogonality theorem: $\theta=0$ in statics $\Rightarrow$ the regulator family is invisible to every tested result	Grounded
Gravity	KM-II	Clustering dial = the soft spot (one dial sets degeneracy-cure + Cerenkov + clusters/CMB)	Grounded
Gravity	KM-II	One-Lambda thesis: $W0 = -24 \cdot \pi^2 \cdot \Omega_{\Lambda} \sim -162$	Grounded
Q-compute	053	$M_{cap} = \min(N_{bw}, N_{V5}, N_{commit})$ is one substrate object; min-structure forced by simultaneity	Grounded
Q-compute	054	The three UR-1 rate quantities have distinct PRIMARY DRIVERS (basis of A/B/C distinctness)	Grounded
Q-compute	055	A/B/C three-class EXHAUSTIVENESS — every architecture saturates one of the three constituents	Grounded

Q-compute	057	Class-B exponential gap-suppression $\Gamma_{\text{fail}} \sim \Gamma_0 e^{-\xi/\xi_0}$	Grounded
Q-compute	058	Class-C plateau $\Gamma_{\text{fail}} \rightarrow \Gamma_{\text{plateau}} > 0$ at redundancy saturation	Grounded
Q-compute	059	Meta-architectures (EC/DD/reservoir/hybrids) = compositions that shift which constituent binds -- NO fourth class	Grounded
Q-compute	060	M_{crit} unification: the matter-wave Q-C boundary and the qubit walls are one substrate threshold	Grounded
QFT	013	V1 QFT-arc propagator (Kallen-Lehmann retarded form)	Grounded
QFT	014	N1+GR1: V1 finite-width + 2-point function on a curved acoustic background	Grounded
QFT	017	Free scalar Klein-Gordon $(\Box + m^2)\phi = 0$ as the DCGT limit of V1	Grounded
QFT	018	Non-abelian YM potential, field strength, YM equation	Grounded
QFT	MS-II	Substrate is a $U(N)$ lattice gauge theory; escapes Nielsen-Ninomiya	Grounded
QFT	MS-II	Spin-1/2 from relationality (tethered node carries $SU(2) \rightarrow SO(3)$ double cover)	Grounded
QFT	MS-II	Parity violation is non-abelian: NO parity-violating abelian force	Grounded
QFT	MS-II	Chirality γ^5 = the arrow's 'screw'	Grounded
QFT	MS-II	$\{1,2,3\} \Leftrightarrow$ internal amplitude dimension $d=3$ (Welch-bound); no stable $SU(N \geq 4)$	Grounded
QFT	MS-II	Three spatial dimensions because the arrow must hold committed order	Grounded
QM-kinematics	003	Born rule $f_K = c_K ^2$ as the participation-frequency limit	Grounded
QM-kinematics	003.5	$P_K = \sqrt{b_K} \cdot e^{i\pi_K}$ amplitude carrier, complex-valued forced	Grounded
QM-kinematics	004.5	Sesquilinear inner-product form $\langle P, Q \rangle = \sum P_K^* Q_K$ (discrete)	Grounded
QM-kinematics	004.6	Continuum inner-product $\int \Psi^* \Phi \, d\mu$ on $L^2(M, d\mu)$	Grounded
QM-kinematics	005	Projective measurement (projector P_K , collapse, Born prob)	Grounded
QM-kinematics	005.5	Double-slit interference $\rho = \sum P_K ^2$; which-path = V5 injection	Grounded
QM-kinematics	006	Unitary evolution $\rightarrow$ Schrodinger $i\hbar \partial_t \Psi = H \Psi$	Grounded
QM-kinematics	006.5	Schrodinger equation via Stone's theorem	Grounded
QM-kinematics	006.6	Kinetic operator $T = p^2/2m$, the factor-2	Grounded
QM-kinematics	006.7	Schrodinger emergence in the thin-participation limit	Grounded
QM-kinematics	007	Hilbert space = Cauchy completion of the motif algebra	Grounded

QM-kinematics	008	Phase structure $S^1/U(1)$ on rule-types	Grounded
QM-kinematics	008.5	Phase-independence of bandwidth = $U(1)$ gauge invariance	Grounded
QM-kinematics	012.5	Momentum operator $p = -i \hbar \text{grad}$ = translation generator	Grounded
QM-kinematics	012.6	Heisenberg $Dx \cdot Dp \geq \hbar/2$	Grounded
Relativistic-QM	102	Spin-statistics $\eta = (-1)^{2s}$	Grounded
Relativistic-QM	106 (T4)	Dirac equation $(i \gamma^\mu d_\mu - mc/\hbar)\Psi = 0$	Grounded
Relativistic-QM	107	Klein-Gordon equation $(\Box + m^2 c^2/\hbar^2)\Psi = 0$	Grounded
Relativistic-QM	108	Minimal-coupling KG + conserved 4-current	Grounded
Relativistic-QM	109	Lorentz-representation taxonomy	Grounded
Relativistic-QM	111	Primitive-level UV finiteness	Grounded
Relativistic-QM	112 (T10)	Lightlike worldlines for $\sigma = 0$ rule-types	Grounded
Relativistic-QM	113	Mass structural form (Arc-M H1)	Grounded
Relativistic-QM	114 (GRH-D1)	Existence of ≥ 1 massless Case-P gauge rule-type	Grounded
Relativistic-QM	115	Vacuum + particle dual Fock structure ($\sigma = 0$)	Grounded
Relativistic-QM	116	MR-R / MR-P conditional massless slot	Grounded
State-reduction	ChannelGranularity	ED's channel/pointer basis is COARSE ($\gg I_{ED}$), so bound charges are single-channel	Grounded
State-reduction	CollapseRate	$\tau \sim \hbar/E_G$ SCALING from ED's primitives (two branch commitment-clocks $N \sim \sqrt{b}$ the single arrow cannot hold $\rightarrow$ einselection individuates)	Grounded
State-reduction	CollapseRate	Two refinements over the rivals: a DEFINITE reduction (not Diosi's mere decoherence) + a NATIVE regulator for the point-mass divergence	Grounded
State-reduction	CollapseRate	NOT excluded by the 2020 spontaneous-radiation bound (Lindblad jump operators = channel projectors $J = g \cdot \Pi$, FORCED from P11; committed states = exact fixed points $D[\Pi] = 0$; no continuous diffusion)	Grounded
State-reduction	OpenPiece-i	MOND enhancement of E_G (mesoscopic masses sub- a_0 in self-gravity $\rightarrow$ naive $\sim 5\text{-}50x$) is CLOSED by ED's own External Field Effect (Paper_029 horizon dipole sourced by total accel; Earth Newtonizes to $\sim 3.5e-6 \rightarrow$ ED $\sim$ Penrose-Diosi in the lab)	Grounded
State-reduction	Positioning §3.2/§6	Locality WITHOUT superdeterminism (local commitment + A1 capacity-zero + stochastic-at-commitment; no hidden variable)	Grounded
Substrate-evaluation	A2 (GrownNotInstalled)	A grown decoupling boundary reproduces exact channel-zero	Grounded

Substrate-evaluation	B4/Maxwell	Integral Gauss law circulation= $2 \cdot \pi \cdot w$ (loop-independent); the Maxwell action is LATENT in the coherence deficit $(\sin^2(d\phi/2) \sim 1/4 (\text{grad } \phi)^2)$	Grounded
Substrate-evaluation	CleanSubstrateVector	$\gamma^5 = \text{arrow} \times \text{spontaneous spatial orientation}$; the vector clean substrate is automatically anomaly-free	Grounded
Substrate-evaluation	GaugeStructure	Gauge group $U(N) = SU(N) \times U(1)/Z_N$ form; F^2 action; mass-gap MECHANISM [A,A] lifts flat direction iff non-abelian; single $U(1)_Y$	Grounded
Substrate-evaluation	Gleason keystone	Channel orthogonality REDUCES to channel-distinctness given the representation; einselection primitive; Born non-contextuality; C selected	Grounded
Substrate-evaluation	V5AttractiveSign/PhaseCoherence	V5's coupling IS the coherence content of the cross-chain superposition $(\sqrt{b_A b_B} \cos(d\phi - \text{int } A))$; MOND cross-term constructive (P12-Coh rewards alignment)	Grounded
Substrate-evaluation	V5UnifiedBudget/Quantum Darwinism	$\rho_{\text{loc}} = 1$: monogamy budget = Class-C plateau budget (one W_{max} spent two ways); per-locus live cap $N_{\text{live}} \leq W_{\text{max}}/w_{\text{bar}}$	Grounded
Black-hole	HorizonTiling	Horizon tiles ~ 1 bit/Planck cell (3 counts: holo=1, frozen 0.78, edge 0.88)	Measured
Gravity	GR-III	Dynamical rule $b \cdot \text{grad}^2 b - \kappa \cdot \rho$: steady state $\text{grad}^2 b \sim \rho$ (0.999); $r_s \sim M$; emergent frozen $b > 0$ horizon; area law $S \sim A$	Measured
Gravity	GR-III	Hawking scaling $\kappa = 1/(2 r_s) \sim 1/r_h$	Measured
Gravity	QuadraticStrain	Phase-coherence constructive sign supplied (P12-Coh rewards alignment)	Measured
QFT	MS-II	Linking held in 3D, erasable in 4D	Measured
QFT	MS-II	Retarded ('arrow') hopping produces net chirality signatures	Measured
Substrate-evaluation	AreaLaw/V5UnifiedBudget	Straddling cross-chain (V5) edge count scales as AREA ($r^{2.02}$), reach-independent; holographic entropy = the edge count ($\sim 0.88/\text{Planck cell}$)	Measured
Substrate-evaluation	KineticLatticeGas/CanonicalPDE	Certified continuum = kinetic lattice-gas (BALLISTIC), NOT a diffusion PDE; and does NOT generate the UDM mobility law ($\beta \sim 0$ not 2)	Measured
Substrate-evaluation	KnotsNotCrystals/RelationaITick	No spontaneous long-range correlation-order in any sector; persistence is rhythmic (V5 binds local proper time, preserving time dilation)	Measured
Substrate-evaluation	MassWithoutMass	Binding confines massless fronts into a sub-luminal inertial composite (mass without mass); mass(binding) != time-dilation (commit-rate)	Measured
Substrate-evaluation	Maxwell	The coherence-weighted continuum limit IS Coulomb, but the DETERMINATE substrate does NOT cast the smooth field (emergent shadow)	Measured

Substrate-evaluation	PhaseCoherence/V5AttractiveSign	P12 coherence rewards alignment but stays FINITE-REACH (no crystal) 2D+3D; V5's attractive local-lock from P12's positive sign (flip destroys it)	Measured
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